MILITARY SPECIFICATION

GAGES, SNAP, THREAD ROLL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers basic requirements for the manufacture, inspection, and preparation for delivery of thread roll snap gages and is to be used in conjunction with drawings or documents that depict the type and style of the desired gages.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-515 — Plywood, Container Grade

UU-P-268 — Paper, Kraft, Untreated, Wrapping

PPP-B-585 — Boxes, Wood, Wirebound

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated

PPP-B-601 — Boxes, Wood, Cleated-Plywood

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner

PPP-B-636 — Boxes, Fiberboard

PPP-T-76 — Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For carton Sealing)

MILITARY

MIL-P-116 — Preservation, Methods of

MIL-B-121 — Barrier Material, Greaseproofed, Waterproofed, Flexible

MIL-S-45176 — Specimens, Comparison, Surface Roughness (Flat) with Carrying Case

MIL-I-45177 — Instruments, Tracer, Surface Roughness

STANDARDS

MILITARY

MIL-STD-10 — Surface Roughness Waviness and Lay

MIL-STD-120 — Gage Inspection

MIL-STD-129 — Marking for Shipment and Storage

MIL-STD-171 (Ord) — Finishing of Metal and Wood Surfaces

FSC 5220

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PUBLICATIONS

U. S. DEPARTMENT OF COMMERCE,
NATIONAL BUREAU OF STANDARDS

Handbook H28 "Screw-Thread Standards for Federal Services"

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Frame

3.1.1 The frame shall be of such size and shape most suitable for the intended purpose. Unless otherwise specified the material for the frame shall be a fine grained cast iron of uniform solidity and density free from all imperfections affecting serviceability or usefulness. The frame shall have features that will permit the thread rolls to have movement, without shake, of the minimum amount specified on applicable documents in a plane perpendicular to the axes of the thread rolls when the thread rolls are assembled on the frame. These features shall have an adjusting and locking device to prevent further movement of the thread rolls once they have been set to the desired specified size. The adjusting and locking device shall be designed so that it can be sealed with wax or other sealing materials. The material for these features shall be of a high quality electric furnace alloy steel, alloy tool steel, or carbon tool steel and shall, unless other wise specified, have a minimum hardness equivalent to C50 Rockwell and a surface Roughness Height Rating of 32 Max.

3.1.2 *Protective coating.* A protective coating shall be applied to the entire frame, except for the machined surfaces adjacent to the roll mounting, as a preservative measure. The processing as specified on applicable documents shall be in accordance with the requirements as specified in Standard MIL–STD–171(Ord). Unless otherwise specified, the top protective coating shall be the manufacturer's standard color. Frames which are made from material which is inherently noncorrosive or tarnish proof do not require an additional finish or coating.

3.1.3 *Marking disc.* A disc of soft material, preferably brass, shall be fastened by screws or friction to the frame for marking purposes.

3.1.4 *Marking.* The complete thread designation, basic pitch diameter, part, or drawing number shall be etched or stamped on the marking disc. All marking shall be permanent and legible and of suitable height commensurate with the size of the frame. Fractional marking except for the number sizes, is permitted in lieu of the decimal method of designation. If the gage size is marked in decimals, when marking the thread designation, there need only be sufficient decimal places to define the size. Example: .19, .5, .625, 1, 2.5, 3.75, etc.

3.2 Rolls.

3.2.1 *Material.* Unless otherwise specified, the rolls shall be of a high quality, electric furnace alloy steel, alloy tool steel, or carbon tool steel and shall have a minimum hardness equivalent to C60 Rockwell.

3.2.2 *Rib form.*

3.2.2.1 *Go rolls.*

3.2.2.1.1 Unless otherwise specified ribs shall occupy the full length of each roll with alternate ribs removed and unless otherwise specified ribs on rolls of 6 threads per inch and coarser shall be single spaced.

3.2.2.2 *Not go rolls.*

3.2.2.2.1 Each roll shall have two ribs double spaced (alternate rib removed) located, unless otherwise specified, as near either end as practicable.

3.2.3 *Tolerances.* Tolerances on half angle

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and rib spacing shall conform to class X tolerances as specified in Handbook H28.

3.2.4 *Rib flanks.*

3.2.4.1 The rib flanks shall be straight and at least extend to the root clearance specified on applicable documents.

3.2.5 *Finish of flank of ribs.* The flanks of ribs shall be free from amorphous or smear metal.

3.2.6 *Roughness height ratings.* Numerical values denoted on documents or drawings for gaging and other surfaces of the gages are Roughness Height Ratings as described in Standard MIL–STD–10.

3.2.7 *Pitch diameter.* The pitch diameter of the rolls, in contrast to the pitch diameter of the assembled gage, is a calculated basic dimension for design purposes. The pitch diameter of the asembled gage is the distance from a point on the pitch diameter of one roll to a similar point on the pitch diameter of the other roll and is the gaging pitch diameter.

3.2.8 *Variations of pitch diameter.* Variations of pitch diameter of the roll (see 3.2.7) and major diameter from a true cylindrical form, such as taper or out-of-roundness over the entire length of the roll, shall not exceed one-half of the tolerance as specified on the applicable documents. The pitch diameter, major diameter, and center-hole diameter must be concentric within .0002 TIR.

3.2.9 *Setting to size.*

3.2.9.1 Thread roll snap gages shall be set to size by use of truncated thread setting plug gages, thread setting bars or other established methods.

3.2.10 *Workmanship.* Workmanship shall be such that the thread roll snap gages specified herein shall comply with the requirements in section 3. All parts and the final assembly shall comply with the provisions set forth in section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Acceptance inspection.** Acceptance inspection by the Government shall consist either of an examination of the inspection record of each gage as maintained by the supplier or actual inspection (one hundred percent inspection) of each gage or a combination of both methods. Unless otherwise specified, acceptance inspection shall be made at the place of delivery.

4.2.1 *Rejections.* Rejections shall be on an individual basis.

4.3 **Hardness test.** Hardness tests shall be made on standard type hardness testers and shall be made as near the actual roll wear surface as practicable.

4.4 **Surface roughness of gaging surfaces.** Surface roughness shall be determined by an instrument conforming to the requirements of Specification MIL–I–45177. Areas where measurement of surface roughness is difficult or impracticable determination of the surface roughness height value shall be made by visual comparison with type I specimens conforming to Specification MIL–S–45176.

4.4.1 Gaging surfaces shall be visually in-

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spected for seams, cracks, nicks, scratches, rust spots, and other defects which may materially effect the use, durability, or stability of the thread roll snap gage.

4.5 Dimensional.

4.5.1 *Gage frame.* The gage frame complete with its component parts shall be inspected for conformance to the requirements as specified herein and on applicable documents.

4.5.1.1 *Gage Rolls.* The gage rolls shall be inspected by the methods specified in Standard MIL–STD–120 for angle, annular rib spacing, thread height, etc.

4.5.1.2 *Assembled gage.* The assembled gage shall be checked at a temperature of 68 degrees F. to insure that it is functionally correct and can be set to the desired pitch diameter. The parallelism of the pitch and minor diameters shall be inspected by the methods described in Standard MIL–STD–120 and shall be within the tolerances specified on applicable documents.

4.5.1.2.1 The locking mechanism shall be inspected to insure a secure setting.

4.5.1.2.2 The thread rolls shall be inspected to insure free rotation.

4.6 The preservation, packaging, and packof the gages shall be in accordance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A*

5.1.1.1 *Cleaning.* The gages shall be cleaned in accordance with process C–1 of Specification MIL–P–116.

5.1.1.2 *Drying.* The gages shall be dried in accordance with any applicable drying procedure specified in Specification MIL–P–116.

5.1.1.3 *Preservation application.* The gages shall be preserved with type P-10 preservative as specified in Specification MIL–P–116.

5.1.1.4 *Unit packaging.* Each gage shall be packaged by method 1A–15 as specified in specification MIL–P–116 and herein. The gage shall be wrapped with type II, grade A, class 2, barrier material conforming to Specification MIL–B–121 and secured with tape conforming to Specification PPP–T–76. The MIL–B–121 barrier material shall be overwrapped with Kraft paper conforming to Specification UU–P–268.

5.1.2 *Level C.* The gages shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The gages (gages of the same size frame) shall be packed in a close-fitting wire bound box, wood cleated fiberboard box, wood cleated-plywood box, nail wood box, or a fiberboard box conforming to Specification PPP–B–585, class 3, PPP–B–591 (Overseas type), PPP–B–601, style A or B using plywood conforming to Specification NN–P–515, type II or III, class 2, PPP–B–621, class 2, and PPP–B–636, class 2 or 3, respectively. Closure and strapping requirements shall be in accordance with the applicable box specification or appendix thereto. The maximum gross weight of wood or woodcleated boxes shall be 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specifications.

5.2.2 *Level B.* The gages (gages of the same size frame) shall be packed in close-fitting wire bound wood box, wood-cleated fiberboard box, wood cleated-plywood box,

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nail wood box or a fiberboard box conforming to Specification PPP-B-585, class 1, PPP-B-591 (Domestic type), PPP-B-601, (Domestic type), PPP-B-621, class 1, and PPP-B-636, class 1, respectively. Closure shall be as specified in the applicable box specification or appendix thereto. The maximum gross weight of wood or wood-cleated boxes shall be 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 *Level C.* The gages shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking. Unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Intended use. Gages described herein are to be used for acceptance inspection of externally threaded components by Government facilities when applicable.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of gages required.
(c) Level of preservation, packaging, and packing required.
(d) Type of setting gage (see 3.2.9.1).
(e) Availability of inspection records (see 4.1).
(f) Applicable drawings or documents.

Custodians:
Army—MU
Navy—Weps
Air Force—MAA

Preparing activity:
Army—MU

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